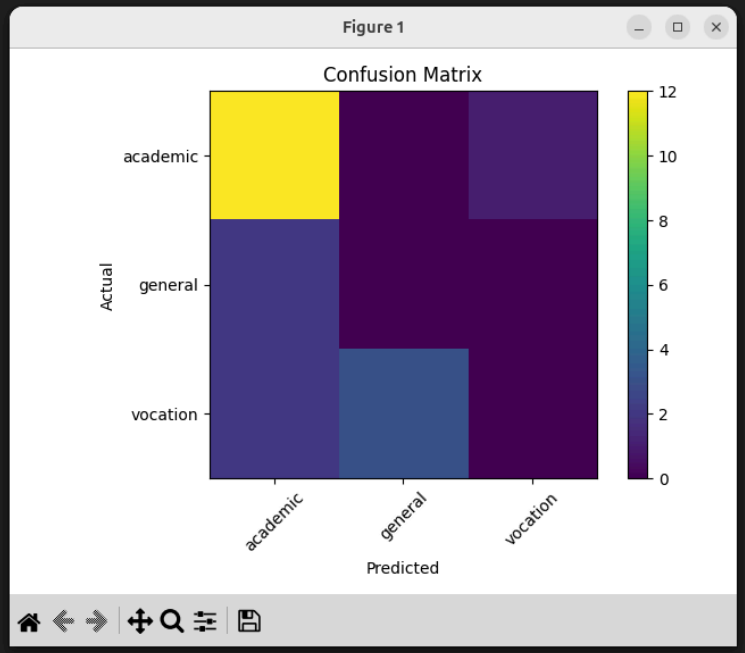
rminal Help

MachineLearning

exercise2.py U | image.png U | exercise1.py U

HW > HW2 > WilliamsBenjamin_HW2 > Exercise1 > exercise1.py > ...

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import matplotlib.pyplot as plt

df = pd.read_csv("hsbdemo.csv")

X = df.drop(columns=["id", "prog", "cid"])
y = df["prog"]

cat_cols = X.select_dtypes(exclude=["number", "bool"]).columns
X = pd.get_dummies(X, columns=cat_cols, drop_first=False)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.10, random_state=9
)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

labels = sorted(y.unique())
cm = confusion_matrix(y_test, y_pred, labels=labels)

print("Misclassified (predicted, actual):")
for i in range(len(y_pred)):
    if y_pred[i] != y_test.iloc[i]:
        print(f"{y_pred[i]}, {y_test.iloc[i]}")

plt.imshow(cm)
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.xticks(range(len(labels)), labels, rotation=45)
plt.yticks(range(len(labels)), labels)
plt.colorbar()
plt.tight_layout()
plt.show()
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW2/WilliamsBenjamin_HW2/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW2/WilliamsBenj
ercise2/exercise2.py
KeyboardInterrupt

ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW2/WilliamsBenjamin_HW2/Exercise2$ cd ../Exercise1
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW2/WilliamsBenjamin_HW2/Exercise1$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW2/WilliamsBenj
ercise1/exercise1.py
Accuracy: 0.6
Misclassified (predicted, actual):
general, vocation
academic, vocation
vocation, academic
general, vocation
academic, vocation
general, vocation
academic, general
academic, general
```

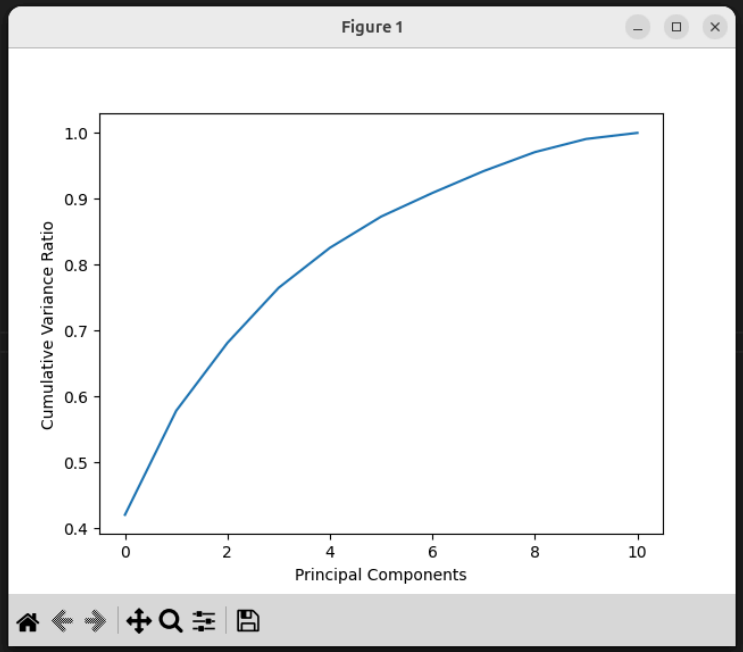
rminal Help

MachineLearning

exercise2.py U | exercise1.py U

HW > HW2 > WilliamsBenjamin_HW2 > Exercise2 > exercise2.py > ...

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

df = pd.read_csv("hsbdemo.csv")

X = df.drop(columns=["id", "prog", "cid"])
y = df["prog"]

cat_cols = X.select_dtypes(exclude=["number", "bool"]).columns
X = pd.get_dummies(X, columns=cat_cols, drop_first=True)

X_scaled = StandardScaler().fit_transform(X)

pca = PCA()
pca.fit(X_scaled)

vr = pca.explained_variance_ratio_
print("Variance Ratios:")

for i in range(len(vr)):
    print(f"PC{i}: {vr[i]:.4f}")

plt.plot(np.cumsum(vr))
plt.xlabel("Principal Components")
plt.ylabel("Cumulative Variance Ratio")
plt.show()
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW2/WilliamsBenjamin_HW2/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW2/WilliamsBenj
ercise2/exercise2.py

ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW2/WilliamsBenjamin_HW2/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW2/WilliamsBenj
ercise2/exercise2.py
Variance Ratios:
PC0: 0.4198
PC1: 0.1581
PC2: 0.1032
PC3: 0.0837
PC4: 0.0607
PC5: 0.0474
PC6: 0.0359
PC7: 0.0332
PC8: 0.0288
PC9: 0.0199
PC10: 0.0092
```